PRESYOPH: A TIME SERIES INFLATION AND HOUSEHOLD PURCHASING POWER ANALYTICS SYSTEM

A Project Proposal with Preliminary Results

Presented for Elective 4 special topics on Data Analytics 2

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Problem Definition

Rising prices are one of the most immediate economic concerns of Filipino households, yet the information needed to judge their effect is difficult for ordinary people to use. Computed from the national Consumer Price Index (CPI) series published by the Philippine Statistics Authority (PSA, 2026), headline inflation reached 6.1% in August 2026, meaning that prices were about 6.1% higher than a year earlier. This increase is not uniform. Price changes differ across the country's 17 regions and across the 13 major commodity groups that make up the CPI, so a single national figure does not describe what a particular household in a particular region actually experiences.

A second gap lies between how often prices and incomes are measured. The CPI is published monthly, while household income and expenditure are measured by the Family Income and Expenditure Survey (FIES), which the PSA has conducted once every three years and, starting with the 2023 round, once every two years (PSA, 2024). Because of this mismatch, a household cannot easily tell at any given moment whether its income is keeping pace with the cost of goods, even though both datasets are publicly available.

This situation affects three groups. Filipino households cannot see whether the real value of their income is eroding. Students and researchers must download and clean separate tables before they can analyze price trends. Local planners lack an accessible regional view of which commodity groups are driving price pressure and where prices are likely to head next.

Data analytics is an appropriate response because both datasets are structured, historical, and sequential, which is the form of data that time series methods are designed to model and forecast (Hyndman & Athanasopoulos, 2021). By combining time series forecasting with interactive visualization, PresyoPH turns published CPI and income statistics into a web-based

tool that supports decisions on two levels. At the household level, it shows what a given income is actually worth after inflation and how it compares with the regional average. At the planning and research level, it shows which commodity groups and regions face the fastest price increases and projects where the CPI is likely to be 6 to 12 months ahead.

Objectives of the Study

General Objective

The primary objective of this project is to design, develop, and evaluate PresyoPH, a web-based time series analytics system that tracks Philippine inflation, forecasts the Consumer Price Index, and estimates household purchasing power using CPI and income benchmark data.

Specific Objectives

1. To acquire, clean, and integrate the PSA monthly CPI data (January 2018 to August 2026) and the 2023 FIES regional income and expenditure benchmarks into a unified, analysis-ready dataset stored in a relational database.
2. To build and compare Prophet, ARIMA, and SARIMA forecasting models that project CPI values 6 to 12 months ahead for each region and commodity group, with confidence intervals.
3. To develop an interactive web-based dashboard that visualizes price change trends and inflation rates by region and commodity group, including a region-by-commodity comparison of where prices are rising fastest.
4. To build a household purchasing power calculator that compares user income against CPI-adjusted value and regional FIES benchmarks.
5. To evaluate forecasting accuracy using RMSE, MAE, and MAPE on a held-out test period, and to evaluate system usability using the System Usability Scale (SUS).

Data Analytics Technique and Methodology

The central analytical technique is time series forecasting applied to monthly CPI series. Each combination of region and commodity group is a separate series of 104 monthly observations from January 2018 to August 2026. The core question the system answers, where prices are headed, is predictive and numeric. It is therefore not suited to classification, clustering, or association rule mining, which assign labels, group similar records, or find items that occur together instead of projecting future values along a timeline (Hyndman & Athanasopoulos, 2021).

Three models are applied to the same series. Prophet is the primary model. It is an additive model that decomposes a series into trend, yearly seasonality, and holiday effects, and it was designed to give reasonable forecasts with little manual tuning (Taylor & Letham, 2018). ARIMA (autoregressive integrated moving average) serves as a non-seasonal baseline. It describes a series through its own past values and past forecast errors after differencing (Box et al., 2015). SARIMA extends ARIMA with seasonal terms so that repeating yearly patterns are modeled explicitly (Box et al., 2015). Benchmarking a more advanced model against simpler, established ones is standard forecasting practice because it shows whether the added complexity actually improves accuracy (Hyndman & Athanasopoulos, 2021). The ARIMA model uses order (1,1,1) and the SARIMA model uses order (1,1,1) with seasonal order (1,1,1,12).

Using all three models on the same data also explains why this technique suits the problem instead of assuming it. Prophet handles trend and seasonality automatically, ARIMA tests how well a model without seasonality performs, and SARIMA shows how much explicit seasonal modeling contributes. The model with the lowest error on held-out data is the one used for the forecast shown in the system.

The system outputs forecasted CPI values for the next 6 to 12 months for every region and commodity group, together with a 95% confidence interval and the year-over-year inflation rate implied by the forecast. The interval lets users see the range of likely outcomes instead of a single number.

Dataset Feasibility and Ethics

This project uses two public datasets released by the PSA. The first is the Consumer Price Index from the PSA OpenSTAT portal (PSA, 2026), which provides monthly CPI values for the Philippines and its 17 regions across 13 major commodity groups plus an All Items series, from January 2018 to August 2026. After cleaning, this dataset contains 26,208 records. The second is the 2023 Family Income and Expenditure Survey (PSA, 2024), from which the group uses 18 benchmark rows, one for the Philippines and one for each region, giving average annual family income and expenditure. Because both datasets are already available and cleaned into structured files, the group does not need to collect any data of its own.

The CPI is structured time series data, while the FIES benchmark is structured cross-sectional data. The two are related through a shared region code. The CPI supplies the monthly backbone for trend analysis and forecasting, and the FIES supplies a regional benchmark against which a user's income is compared.

The CPI dataset is sufficient for the chosen technique. Each series covers more than eight years of consecutive monthly observations, enough to include several full yearly cycles for the seasonal model and to reserve the most recent eight months for testing. The FIES dataset is small on purpose because it is used only as a benchmark and not for model training. Its limitation is timing. The 2023 survey does not reflect income changes since then, and because the PSA now

conducts the FIES every two years, newer rounds will eventually replace it. The group treats this as a known limitation, and the calculator states the survey year next to the benchmark.

Validating the data revealed one quality problem. The Financial Services series is unchanged in every region and nationally: the national CPI has stayed at 142.4 since January 2024, and the regional series have been flat for between 32 and 65 months. One further series (Information and Communication in Region III) has been flat for six months. A CPI that does not move for this long almost certainly reflects a stale or carried-forward value in the published table and not genuinely stable prices. The system therefore detects any series with an unchanged CPI for six or more months, leaves it out of rankings, maps and forecasts, and tells the user why. This affects 19 of the 252 series. The group recommends verifying these values against the PSA release before the final defense.

No privacy or ethical concern is expected. Both datasets are aggregate statistics published by a government agency and contain no information about individual persons or households. The income entered in the calculator is used only for the computation shown on the screen and is not saved in the database. The calculator compares simulated or user-entered income with regional averages and does not use individual household survey records.

Conceptual Framework

The operational architecture follows a sequential pipeline: Input Data > Data Preparation > Analytics Engine > System Output.

Monthly CPI records and the FIES regional benchmarks are loaded, validated, cleaned, joined by region code, and stored in a SQLite database. The database is rebuilt automatically whenever the source files change, so the stored tables always match the published data. The prepared CPI series are passed to the analytics engine, where Prophet, ARIMA, and SARIMA

are fitted on the same history and evaluated on the held-out period, and the best-performing model produces the forecast. When a user enters a monthly income, the system divides it by the latest CPI of the chosen region and multiplies the result by 100 to express it in 2018 base pesos. The outputs are the price change dashboard, the CPI forecast with its confidence interval, and the purchasing power comparison, all displayed in the web dashboard.

System Features

The project deliverable is a functional web system, not only a forecasting model. It serves households, students and researchers, and local planners through four connected features that share one region selector, plus a methodology page that documents the data and formulas.

Price Change Dashboard: Opens with a one-sentence summary generated from the data and four headline figures. A dumbbell chart ranks the 13 commodity groups by year-over-year price change for the selected region and year, marked as normal (3% or lower), increasing (above 3% to 5%), or high (above 5%) inflation, and shows each group against the national figure. A trend view shows headline inflation, region versus national, with the inflation bands shaded, and a small chart for every commodity group. A regional view shows a schematic tile map and a heatmap of the year-over-year rate for every region and commodity group, so planners can see where price pressure is concentrated.

Forecasting Module: Trains the three models for a selected region and commodity group and displays a 6 to 12 month forecast from the best model with its confidence band, a marker at the forecast cutoff, and the year-over-year inflation the forecast implies. A model comparison view reports each model's accuracy on the test period and plots its predictions against the actual values for the months it had not seen. The forecast can be downloaded as a CSV file, and series with a frozen CPI are not forecast.

Inflation Rate Viewer: Shows month-over-month or year-over-year inflation for any region and commodity group as a trend line with the 3% and 5% thresholds shaded, together with the latest reading, the change since the previous reading, and the highest rate recorded since 2018. A calendar view of month-over-month changes shows seasonal patterns.

Purchasing Power Calculator: Accepts a monthly household income, converts it to 2018 base pesos using the latest CPI of the region, shows what one peso still buys compared with 2018, the income that would be needed today to match the buying power of the entered income in 2018, and how the value of a fixed income has changed month by month. It compares the income with the region's average monthly income and expenditure from the FIES, converted from annual figures, and states the survey year.

Evaluation Plan

Forecast accuracy is evaluated by training each model on January 2018 to December 2025 (96 months) and testing it on the held-out period of January to August 2026 (8 months). Three error measures are reported for every model. Root mean squared error (RMSE) penalizes large misses, mean absolute error (MAE) is the average size of the errors in index points, and mean absolute percentage error (MAPE) expresses the error relative to the actual value (Hyndman & Koehler, 2006). The model with the lowest RMSE on the test period is selected for the forecast shown in the dashboard. To judge the models beyond a single series, the same procedure is run in batch on every region and commodity-group series that passes the data-quality check described below (233 of 252 series).

System testing combines functional verification with basic user acceptance testing. Functional testing checks that the dashboard charts match the underlying data and that the calculator returns correct results, including edge cases such as zero income and very large

income values. These checks are automated in a test suite that also covers the database loading, the inflation formulas, the forecasting pipeline, and a smoke test of every page, and it runs on every code change. Usability will be assessed by asking sample users to complete simple tasks, such as finding the fastest-rising category in their region and checking what their income is worth, and then answering the 10-item System Usability Scale (Brooke, 1996). The scores will be interpreted against published benchmarks (Bangor et al., 2008).

Preliminary Results

The batch evaluation scored all 233 series that passed the data-quality check, with no model failures. Nineteen of the 252 series were left out because their CPI had not changed for at least six months (see Dataset Feasibility and Ethics), and scoring a frozen series would flatter every model. Table 1 summarizes accuracy on the January to August 2026 test period. SARIMA had the lowest error on every measure, with a mean RMSE of 3.46 index points, a mean MAE of 3.00, and a mean MAPE of 2.12%. ARIMA had a mean RMSE of 4.40 and Prophet 4.50. Because each series is scored separately and the best model is chosen per series, SARIMA was selected for 147 of the 233 series (63.1%), Prophet for 56 (24.0%), and ARIMA for 30 (12.9%).

Table 1

Forecast Accuracy on the Held-Out Test Period (January to August 2026)

Model	M RMSE	Mdn RMSE	M MAE	M MAPE (%)	Mdn MAPE (%)	Selected, n (%)
ARIMA	4.40	2.81	3.86	2.70	1.85	30 (12.9)
Prophet	4.50	2.71	4.02	2.84	1.79	56 (24.0)
SARIMA	3.46	2.17	3.00	2.12	1.36	147 (63.1)

Note. N = 233 series (18 regions by 14 commodity series, less 19 series with a frozen CPI). M = mean, Mdn = median. RMSE and MAE are in CPI index points. Selected = the model with the lowest test RMSE for a series.

The differences are clearest for the headline All Items series. Across the 18 regional series, SARIMA had a mean RMSE of 3.61, compared with 4.45 for ARIMA and 6.05 for Prophet, and it was selected for 16 of them, with ARIMA selected for the other two. For the national series, SARIMA reached an RMSE of 3.07 and a MAPE of 1.94%, against 3.45 and 2.18% for ARIMA and 5.01 and 3.31% for Prophet. The share of series forecast with a MAPE under 5% was 89.3% for SARIMA, 85.8% for ARIMA, and 82.8% for Prophet, and the largest single MAPE was 10.5% for SARIMA, 12.4% for ARIMA, and 13.1% for Prophet.

Two points temper these results. First, the gap between ARIMA and Prophet is small and depends on the measure: Prophet had the higher mean RMSE (4.50 against 4.40) but the lower median RMSE (2.71 against 2.81), so for a typical series the two are close, and the means are pulled up by a few series where they miss badly. Second, Prophet, the proposed primary model, did not outperform SARIMA in this test, and it was less accurate than the simpler ARIMA model on the headline All Items series. This agrees with the general advice to benchmark advanced models against simple ones rather than assume they are better (Hyndman & Athanasopoulos, 2021). It is a finding about this dataset and test window, not a general ranking: eight test months from one split is a short window, and the 2026 price movements it contains may not repeat. The system therefore keeps all three models and chooses among them for each series. System usability has not yet been measured, and the SUS results will be reported after user testing.

Tools, Technologies, and Scope

The system is implemented in Python 3. Pandas and NumPy are used for data cleaning and manipulation, SQLite for storing the cleaned CPI and benchmark tables, Prophet and statsmodels for fitting the Prophet, ARIMA, and SARIMA models, and scikit-learn for computing the error measures. The web interface and the calculator are built with Streamlit, and

the interactive charts are drawn with Plotly. Automated tests are written with pytest. All of these are free, open-source tools with large user communities. The source code is hosted on GitHub.

The project scope covers national and 17-region monthly CPI from January 2018 to August 2026, the 13 commodity groups plus All Items, the 2023 FIES regional benchmarks, forecasts 6 to 12 months ahead, and the user-input purchasing power calculator. It excludes city-, barangay-, and store-level prices, real-time web scraping, actual household income records (simulated or user-entered values are used instead), and monitoring of individual items such as the price of one kilogram of rice.

Limitations

Five limitations apply. The FIES benchmark reflects 2023 incomes and becomes less representative as prices and wages change. Forecasts extrapolate past patterns and cannot anticipate shocks such as supply disruptions or policy changes, so users should read the confidence interval and not only the central line. The accuracy results come from a single eight-month test window. Nineteen series have a frozen CPI that looks like a data error in the published table; they are excluded until the values are verified, so the system gives no ranking or forecast for Financial Services. Finally, the data are regional aggregates, so the system describes typical conditions in a region and not the experience of any one household.

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